

PRACTICAL – 1

'''

Implement depth first search algorithm and Breadth First Search algorithm, Use an undirected graph and develop a recursive algorithm for searching all the vertices of a graph or tree data structure.

'''

```
from collections import deque
```

Function to add edge (undirected)

```
def add_edge(graph, u, v):  
    graph.setdefault(u, []).append(v)  
    graph.setdefault(v, []).append(u)
```

DFS (recursive)

```
def dfs(graph, node, visited):  
    if node not in visited:  
        print(node, end=" ")  
        visited.add(node)  
        for neighbor in graph.get(node, []):  
            dfs(graph, neighbor, visited)
```

BFS

```
def bfs(graph, start):  
    visited = set()  
    queue = deque([start])  
    visited.add(start)  
  
    while queue:  
        node = queue.popleft()  
        print(node, end=" ")  
  
        for neighbor in graph.get(node, []):  
            if neighbor not in visited:  
                visited.add(neighbor)  
                queue.append(neighbor)
```

----- MAIN PROGRAM -----

```
graph = {}
```

```
n = int(input("Enter number of edges: "))
```

```
print("Enter edges (u v):")
```

```
for _ in range(n):  
    u, v = input().split()  
    add_edge(graph, u, v)
```

```
start = input("Enter starting vertex: ")
```

```
print("\nDFS Traversal:")
```

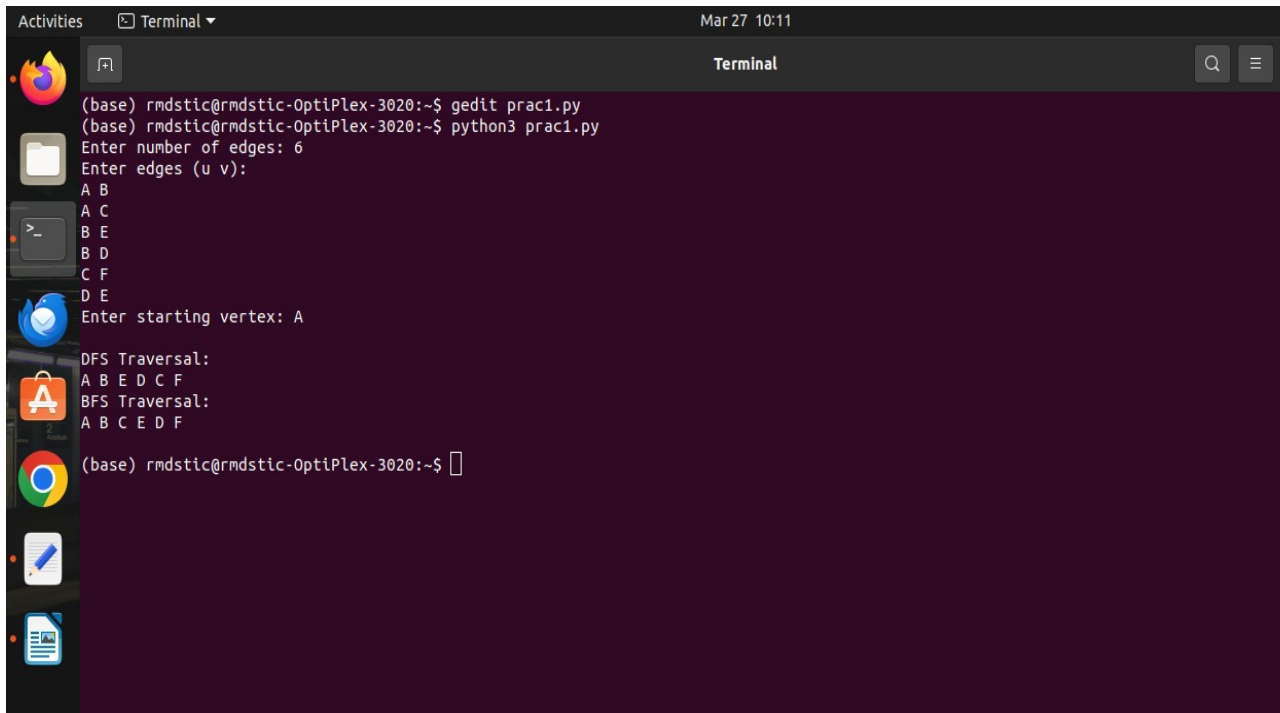
```
dfs(graph, start, set())
```

```
print("\nBFS Traversal:")
```

```
bfs(graph, start)
```

```
print("\n")
```

OUTPUT

A screenshot of a Linux terminal window. The window has a title bar with 'Activities', 'Terminal', and 'Mar 27 10:11'. On the left is a vertical dock with icons for Firefox, Files, Terminal, and others. The terminal itself has a dark purple background. It shows a user running a Python script named 'prac1.py'. The script prompts for the number of edges (6) and then lists six edges: A B, A C, B E, B D, C F, and D E. It then asks for a starting vertex (A) and displays the results of Depth-First Search (DFS) and Breadth-First Search (BFS) traversals, both resulting in the sequence A B E D C F. The prompt returns to the shell.

```
(base) rmdstic@rmdstic-OptiPlex-3020:~$ gedit prac1.py
(base) rmdstic@rmdstic-OptiPlex-3020:~$ python3 prac1.py
Enter number of edges: 6
Enter edges (u v):
A B
A C
B E
B D
C F
D E
Enter starting vertex: A

DFS Traversal:
A B E D C F
BFS Traversal:
A B C E D F

(base) rmdstic@rmdstic-OptiPlex-3020:~$
```